

Traction vs. Drivetrain: An Analysis of Vehicle Safety and Consumer Priorities

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DSC680 Milestone 2

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Introduction

Many consumers believe that all-wheel drive (AWD) is one of the most important factors for maintaining vehicle safety in poor weather conditions. This belief has been reinforced by marketing strategies that position AWD as a premium safety feature, often leading consumers to pay more for vehicles equipped with it. However, while AWD can improve acceleration in low-traction environments, it does not directly improve braking performance or overall grip in the same way that tire quality does.

This project explores whether consumers may be prioritizing drivetrain type over more impactful safety factors. By analyzing accident data, vehicle characteristics, and external research on tire performance, this study aims to better understand how real-world safety outcomes relate to traction conditions and vehicle design.

Data Overview

This analysis uses multiple datasets to examine both safety outcomes and consumer behavior.

The primary dataset is the U.S. Accidents dataset from Kaggle, which includes millions of accident records across the United States. Key variables include accident severity, weather conditions, visibility, and location. This dataset is used to analyze how environmental conditions impact crash outcomes.

A secondary dataset includes vehicle features such as drivetrain type (FWD, AWD, RWD), pricing, and model information. This dataset is used to explore how AWD vehicles are distributed in the market and whether they are associated with higher prices.

Since tire type is not included in either dataset, external sources such as Tire Rack and AAA are used to support conclusions about traction and braking performance.

Methodology

The analysis begins with exploratory data analysis (EDA) to identify patterns in accident severity under different environmental conditions. Weather conditions such as snow, rain, and clear conditions are compared to determine how they influence crash severity.

Comparative analysis is then performed to examine how drivetrain types are distributed across vehicles and whether AWD vehicles are associated with higher pricing. This helps establish the role of AWD as a marketed feature.

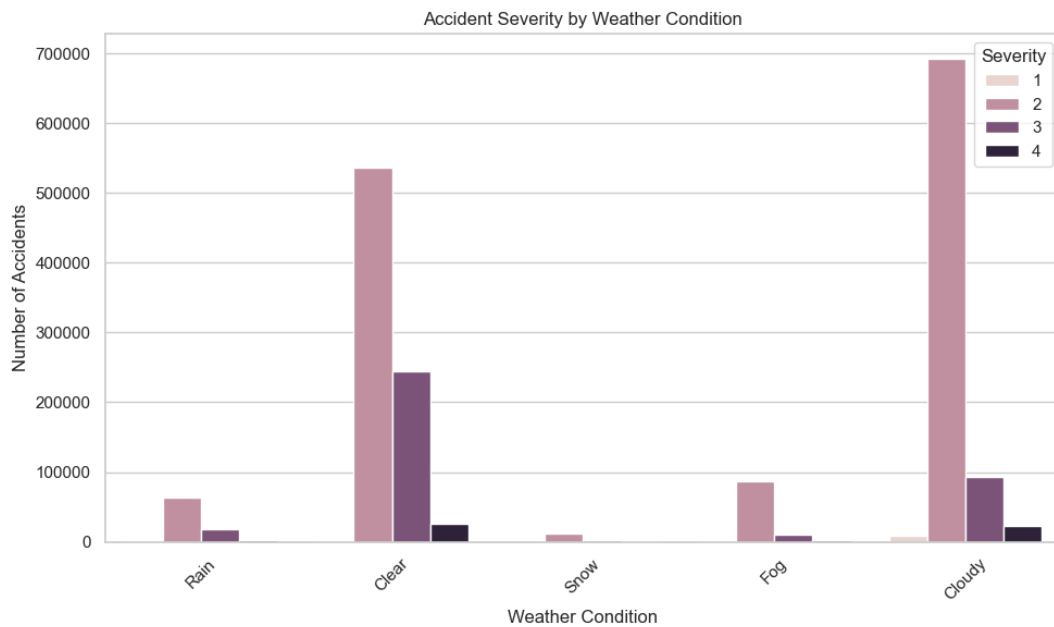
A basic predictive model, such as logistic regression, may be used to estimate the likelihood of severe accidents based on environmental variables like weather and visibility. The goal of this model is not to achieve perfect accuracy, but to support the overall analysis by identifying key contributing factors.

Data visualization is used throughout the project to clearly communicate trends and relationships in a way that is easy to interpret.

Results and Analysis

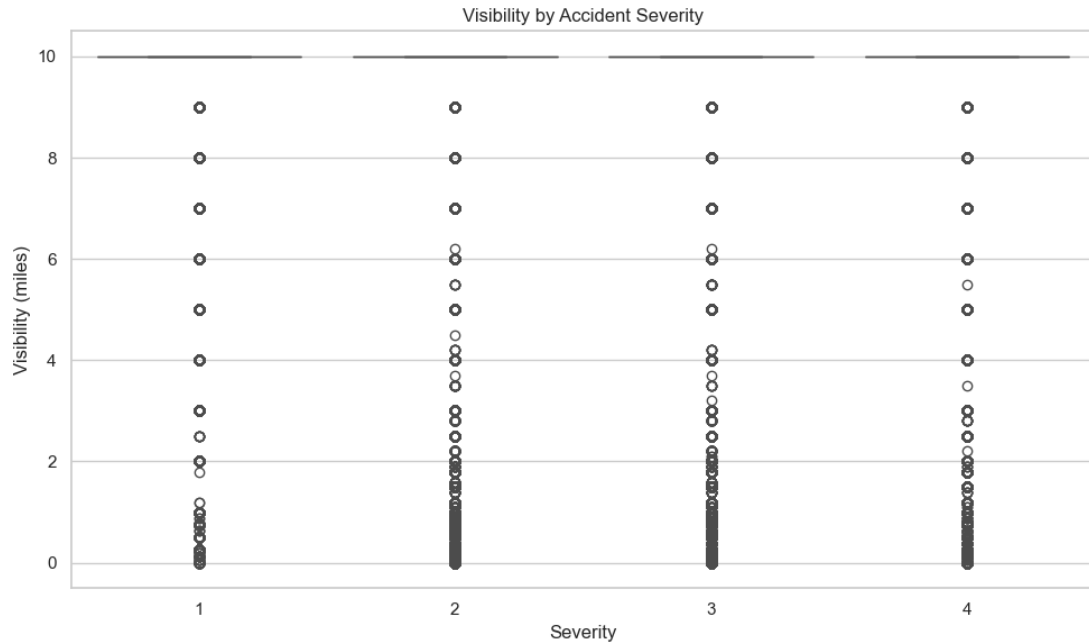
Initial analysis shows that accident severity tends to increase under adverse weather conditions such as snow, ice, and low visibility. This suggests that environmental factors related to traction play a significant role in safety outcomes.

Figure 1: Accident Severity by Weather Condition



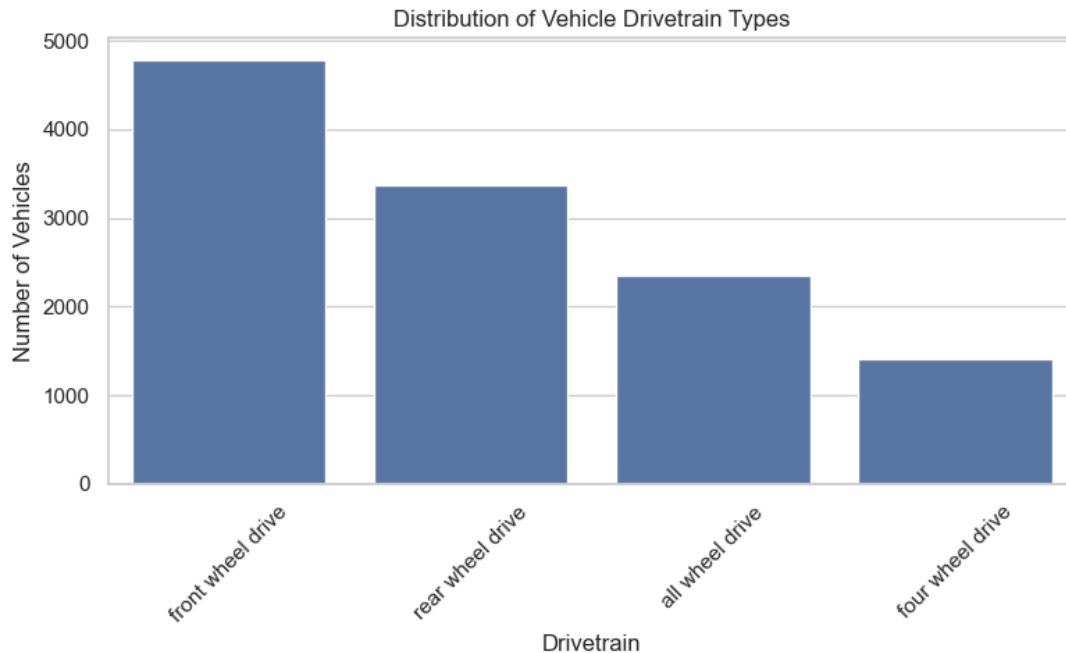
Further analysis of visibility and environmental variables indicates that reduced visibility and road conditions are strongly associated with more severe accidents.

Figure 2: Severity vs Visibility



When examining vehicle data, AWD vehicles are often associated with higher prices and are more common in certain vehicle types such as SUVs. This supports the idea that AWD is positioned as a premium feature in the market.

Figure 3: Drivetrain Distribution



While drivetrain type may influence certain aspects of driving performance, it does not directly address braking distance or tire-road interaction. External research supports the idea that tire performance has a significant impact on traction and stopping ability, particularly in winter conditions.

Discussion

The results suggest that environmental conditions, particularly those affecting traction, are a major factor in accident severity. While AWD provides benefits in acceleration, it does not directly improve braking or overall traction in the same way that tire quality does. This creates a potential mismatch between consumer perception and actual safety outcomes. Consumers may prioritize AWD when purchasing vehicles, while underestimating the importance of tire selection.

It is important to note that this analysis does not directly measure tire type, which limits the ability to draw definitive conclusions. However, by combining real-world accident data with external performance research, a consistent pattern emerges that highlights the importance of traction-related factors.

Conclusion

This project explored the relationship between drivetrain type, environmental conditions, and vehicle safety outcomes. The findings suggest that while AWD is commonly marketed as a safety feature, environmental factors such as weather and visibility have a stronger impact on accident severity.

Although tire data was not directly available, external research indicates that tire performance plays a critical role in traction and braking. This suggests that consumers may benefit from placing greater emphasis on tire selection rather than relying solely on drivetrain features. Future work could include more detailed datasets that incorporate tire type, allowing for a more direct analysis of its impact on safety outcomes.

References

Kaggle. (n.d.). *U.S. accidents dataset (2016–2023)*.

<https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>

Kaggle. (n.d.). *Car features and MSRP dataset*.

<https://www.kaggle.com/datasets/CooperUnion/cardataset>

American Automobile Association. (2019). *Winter driving tips and safety information*.

<https://exchange.aaa.com/safety/driving-advice/winter-driving-tips/>

Tire Rack. (n.d.). *Tire testing and performance comparisons*.

<https://www.tirerack.com/tires/tests/>

Federal Highway Administration. (n.d.). *Road weather management*.

<https://ops.fhwa.dot.gov/weather/>

Appendix

Appendix A – Data Preparation and Processing

The datasets used in this project required preprocessing before analysis. The U.S. Accidents dataset was cleaned to remove missing values and focus on relevant variables such as weather conditions, visibility, severity, and time of occurrence.

The Start_Time column was converted to a datetime format to extract the month and categorize each record into a season. This allowed for seasonal analysis of accident trends.

Questions

1. Does AWD actually improve safety?

AWD improves acceleration in low-traction conditions, but it does not improve braking distance or overall traction, which are critical for avoiding accidents.

2. Why didn't you include tire data directly?

Most large-scale accident datasets do not include tire type. Because of this, external research and testing data were used to support conclusions about tire performance.

3. What was the most important factor affecting accident severity?

Weather conditions and visibility appeared to have the strongest relationship with accident severity.

4. Why did some seasons have more accidents than others?

Seasonal accident counts are influenced by driving behavior and volume. For example, fall may have more total accidents due to increased travel and commuting.

5. Does this mean AWD is not useful?

No, AWD still has benefits, especially for acceleration in poor conditions. However, it should not be considered the most important factor for safety.

6. Could driver behavior have affected the results?

Yes, driver behavior is a major factor in accidents and is not fully captured in the dataset.

7. Why did you use multiple datasets?

No single dataset contained all the variables needed, so multiple datasets were combined to provide a more complete analysis.

8. How accurate is the model you used?

The model was used to identify trends and relationships rather than to make highly accurate predictions.

9. What would improve this analysis?

Access to tire-specific data and maintenance records would allow for a more direct comparison of tire performance and safety outcomes.

10. What is the main takeaway from this project?

Environmental conditions and traction-related factors appear to play a larger role in safety outcomes than drivetrain type alone, suggesting that consumers may benefit from focusing more on tire performance.

DSC680 Ayden Trusty Milestone 2 Jupyter Notebook

April 5, 2026

1 DSC680 Milestone 2

1.1 Ayden Trusty

Datasets:

<https://www.kaggle.com/datasets/CooperUnion/cardataset>

<https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents?>

```
[3]: #Import packages
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
```

```
[10]: import os

os.chdir(r"C:\Users\ayden\Documents\College")
print(os.getcwd())
```

C:\Users\ayden\Documents\College

```
[11]: #Load datasets
accidents = pd.read_csv("US_Accidents_March23.csv")
cars = pd.read_csv("data.csv")
```

```
[12]: print(accidents.columns.tolist())
print(cars.columns.tolist())
```

```
['ID', 'Source', 'Severity', 'Start_Time', 'End_Time', 'Start_Lat', 'Start_Lng',
'End_Lat', 'End_Lng', 'Distance(mi)', 'Description', 'Street', 'City', 'County',
'State', 'Zipcode', 'Country', 'Timezone', 'Airport_Code', 'Weather_Timestamp',
'Temperature(F)', 'Wind_Chill(F)', 'Humidity(%)', 'Pressure(in)',
'Visibility(mi)', 'Wind_Direction', 'Wind_Speed(mph)', 'Precipitation(in)',
'Weather_Condition', 'Amenity', 'Bump', 'Crossing', 'Give_Way', 'Junction',
'No_Exit', 'Railway', 'Roundabout', 'Station', 'Stop', 'Traffic_Calming',
'Traffic_Signal', 'Turning_Loop', 'Sunrise_Sunset', 'Civil_Twilight',
'Nautical_Twilight', 'Astronomical_Twilight']
```

```
['Make', 'Model', 'Year', 'Engine Fuel Type', 'Engine HP', 'Engine Cylinders',
'Transmission Type', 'Driven_Wheels', 'Number of Doors', 'Market Category',
'Vehicle Size', 'Vehicle Style', 'highway MPG', 'city mpg', 'Popularity',
'MSRP']
```

```
[13]: accidents.head()
```

```
[13]:      ID  Source  Severity      Start_Time      End_Time \
0  A-1  Source2         3  2016-02-08 05:46:00  2016-02-08 11:00:00
1  A-2  Source2         2  2016-02-08 06:07:59  2016-02-08 06:37:59
2  A-3  Source2         2  2016-02-08 06:49:27  2016-02-08 07:19:27
3  A-4  Source2         3  2016-02-08 07:23:34  2016-02-08 07:53:34
4  A-5  Source2         2  2016-02-08 07:39:07  2016-02-08 08:09:07

      Start_Lat  Start_Lng  End_Lat  End_Lng  Distance(mi)  ... Roundabout \
0  39.865147 -84.058723      NaN      NaN          0.01  ...      False
1  39.928059 -82.831184      NaN      NaN          0.01  ...      False
2  39.063148 -84.032608      NaN      NaN          0.01  ...      False
3  39.747753 -84.205582      NaN      NaN          0.01  ...      False
4  39.627781 -84.188354      NaN      NaN          0.01  ...      False

      Station  Stop  Traffic_Calming  Traffic_Signal  Turning_Loop  Sunrise_Sunset \
0   False  False          False          False          False          Night
1   False  False          False          False          False          Night
2   False  False          False          True          False          Night
3   False  False          False          False          False          Night
4   False  False          False          True          False          Day

      Civil_Twilight  Nautical_Twilight  Astronomical_Twilight
0              Night              Night              Night
1              Night              Night              Day
2              Night              Day              Day
3              Day              Day              Day
4              Day              Day              Day
```

```
[5 rows x 46 columns]
```

```
[14]: cars.head()
```

```
[14]:      Make      Model  Year      Engine Fuel Type  Engine HP \
0   BMW  1 Series M  2011  premium unleaded (required)    335.0
1   BMW  1 Series  2011  premium unleaded (required)    300.0
2   BMW  1 Series  2011  premium unleaded (required)    300.0
3   BMW  1 Series  2011  premium unleaded (required)    230.0
4   BMW  1 Series  2011  premium unleaded (required)    230.0

      Engine Cylinders  Transmission Type      Driven_Wheels  Number of Doors \
0              6.0          MANUAL  rear wheel drive          2.0
```

1	6.0	MANUAL	rear wheel drive	2.0
2	6.0	MANUAL	rear wheel drive	2.0
3	6.0	MANUAL	rear wheel drive	2.0
4	6.0	MANUAL	rear wheel drive	2.0

	Market Category	Vehicle Size	Vehicle Style	\
0	Factory Tuner,Luxury,High-Performance	Compact	Coupe	
1	Luxury,Performance	Compact	Convertible	
2	Luxury,High-Performance	Compact	Coupe	
3	Luxury,Performance	Compact	Coupe	
4	Luxury	Compact	Convertible	

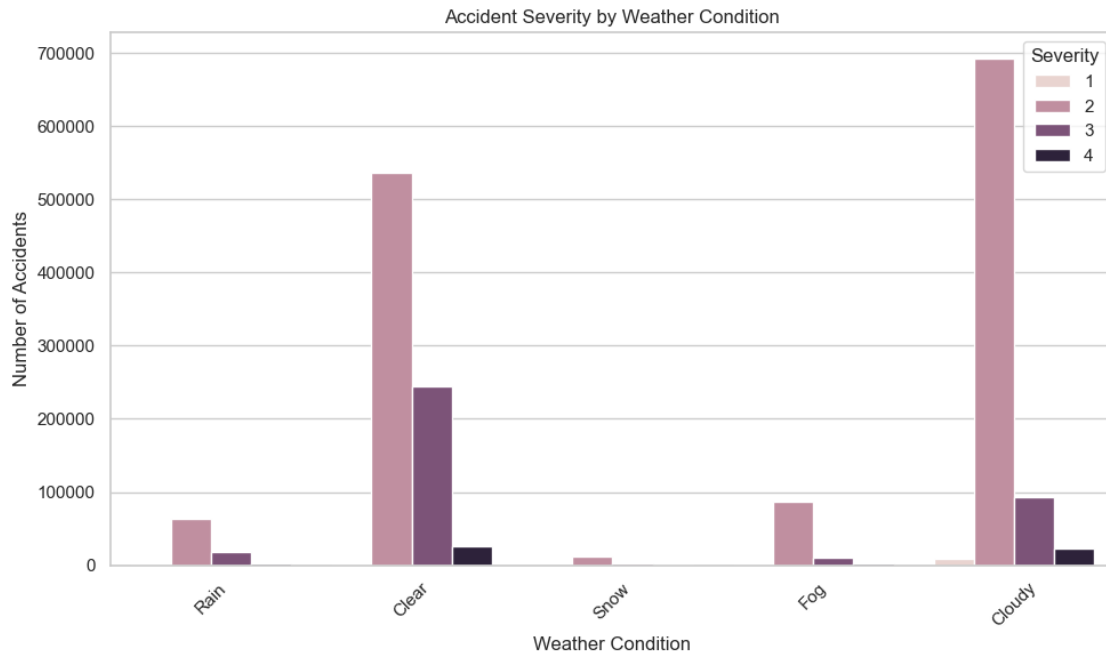
	highway MPG	city mpg	Popularity	MSRP
0	26	19	3916	46135
1	28	19	3916	40650
2	28	20	3916	36350
3	28	18	3916	29450
4	28	18	3916	34500

1.1.1 Accident severity by weather condition

```
[15]: # Keep only the weather conditions I want to compare
weather_subset = accidents[accidents["Weather_Condition"].notna()].copy()

selected_weather = ["Clear", "Rain", "Snow", "Fog", "Cloudy"]
weather_subset = weather_subset[weather_subset["Weather_Condition"].
    ↪isin(selected_weather)]

# Plot
plt.figure(figsize=(10, 6))
sns.countplot(data=weather_subset, x="Weather_Condition", hue="Severity")
plt.title("Accident Severity by Weather Condition")
plt.xlabel("Weather Condition")
plt.ylabel("Number of Accidents")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

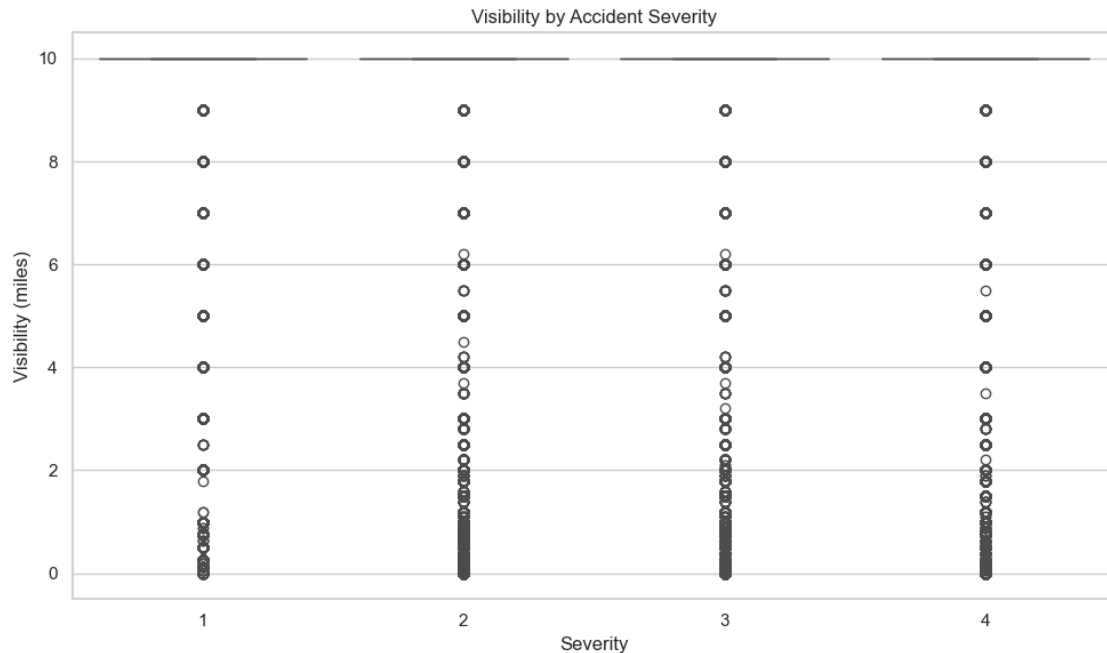


1.1.2 Accident severity by visibility

```
[16]: visibility_subset = accidents[["Visibility(mi)", "Severity"]].dropna().copy()

visibility_subset = visibility_subset[
    (visibility_subset["Visibility(mi)"] >= 0) &
    (visibility_subset["Visibility(mi)"] <= 10)
]

plt.figure(figsize=(10, 6))
sns.boxplot(data=visibility_subset, x="Severity", y="Visibility(mi)")
plt.title("Visibility by Accident Severity")
plt.xlabel("Severity")
plt.ylabel("Visibility (miles)")
plt.tight_layout()
plt.show()
```

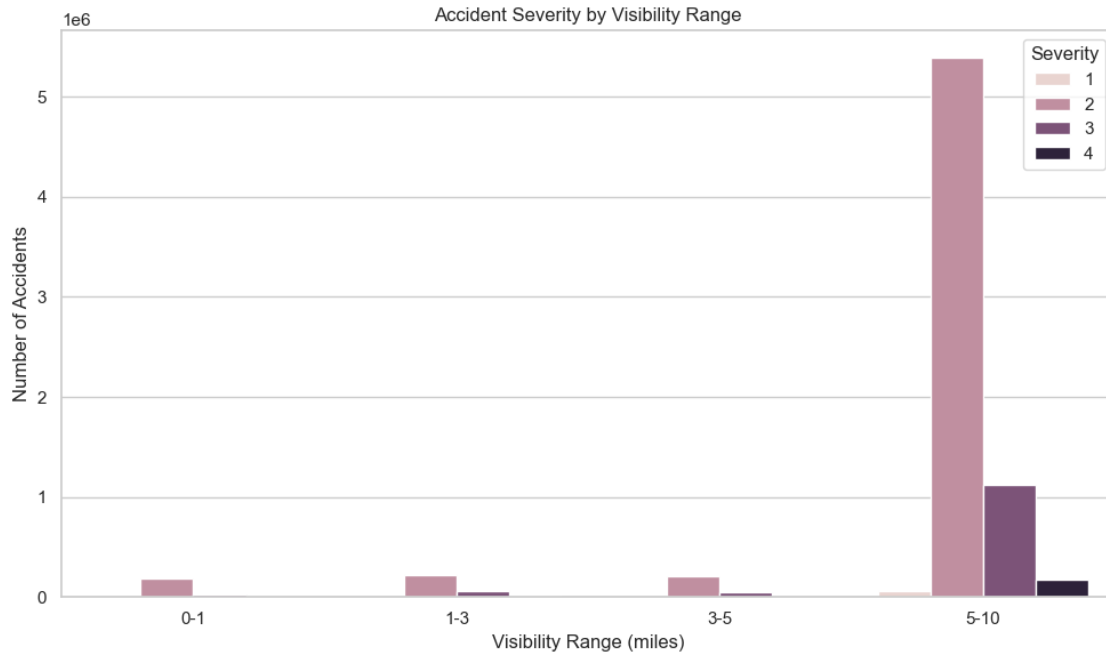



1.1.3 Accident severity by visibility Range

```
[17]: visibility_subset = accidents[["Visibility(mi)", "Severity"]].dropna().copy()
visibility_subset = visibility_subset[
    (visibility_subset["Visibility(mi)"] >= 0) &
    (visibility_subset["Visibility(mi)"] <= 10)
]

bins = [0, 1, 3, 5, 10]
labels = ["0-1", "1-3", "3-5", "5-10"]
visibility_subset["Visibility_Bin"] = pd.cut(
    visibility_subset["Visibility(mi)"], bins=bins, labels=labels,
    include_lowest=True
)

plt.figure(figsize=(10, 6))
sns.countplot(data=visibility_subset, x="Visibility_Bin", hue="Severity")
plt.title("Accident Severity by Visibility Range")
plt.xlabel("Visibility Range (miles)")
plt.ylabel("Number of Accidents")
plt.tight_layout()
plt.show()
```



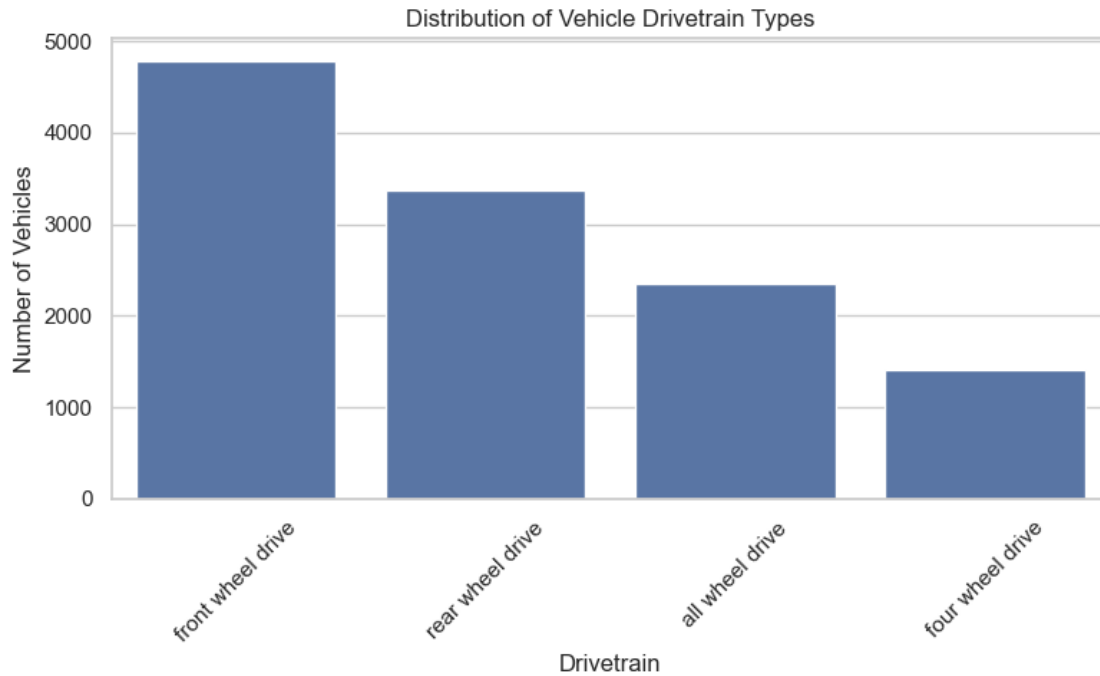
```
[18]: print(cars.columns.tolist())
```

```
['Make', 'Model', 'Year', 'Engine Fuel Type', 'Engine HP', 'Engine Cylinders',
 'Transmission Type', 'Driven_Wheels', 'Number of Doors', 'Market Category',
 'Vehicle Size', 'Vehicle Style', 'highway MPG', 'city mpg', 'Popularity',
 'MSRP']
```

1.1.4 Drivetrain distribution

```
[21]: drive_subset = cars[["Driven_Wheels"]].dropna().copy()

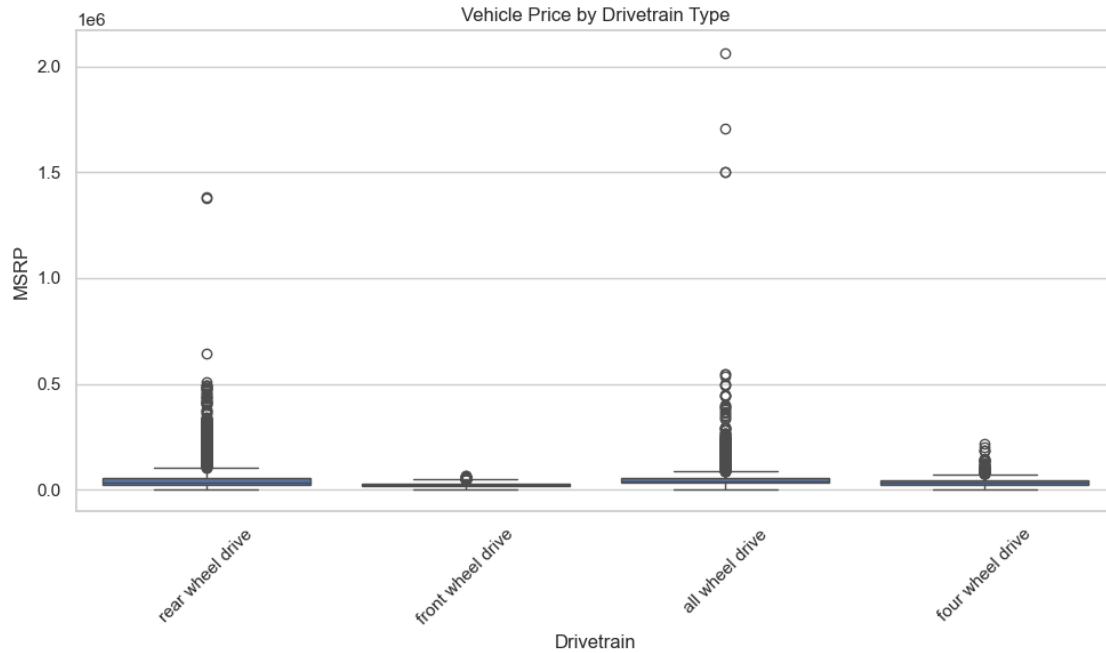
plt.figure(figsize=(8, 5))
sns.countplot(data=drive_subset, x="Driven_Wheels",
              order=drive_subset["Driven_Wheels"].value_counts().index)
plt.title("Distribution of Vehicle Drivetrain Types")
plt.xlabel("Drivetrain")
plt.ylabel("Number of Vehicles")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



1.1.5 Price by drivetrain

```
[23]: price_subset = cars[["Driven_Wheels", "MSRP"]].dropna().copy()

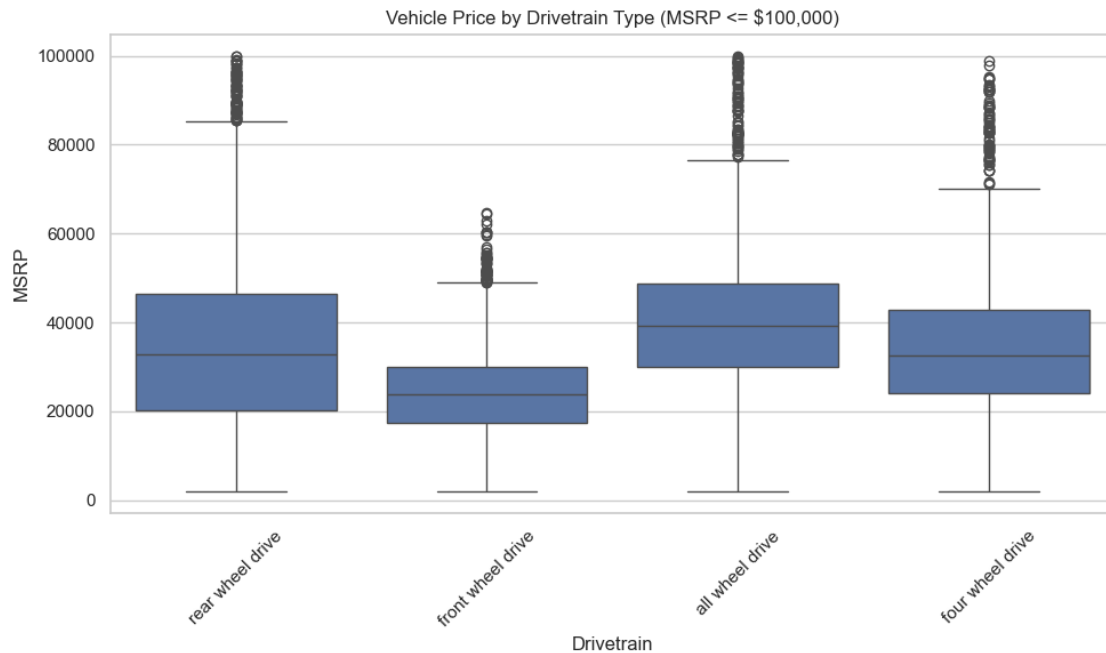
plt.figure(figsize=(10, 6))
sns.boxplot(data=price_subset, x="Driven_Wheels", y="MSRP")
plt.title("Vehicle Price by Drivetrain Type")
plt.xlabel("Drivetrain")
plt.ylabel("MSRP")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



1.1.6 Vehicle Prices by drivetrain under \$100k

```
[25]: price_subset = cars[["Driven_Wheels", "MSRP"]].dropna().copy()
price_subset = price_subset[price_subset["MSRP"] <= 100000]

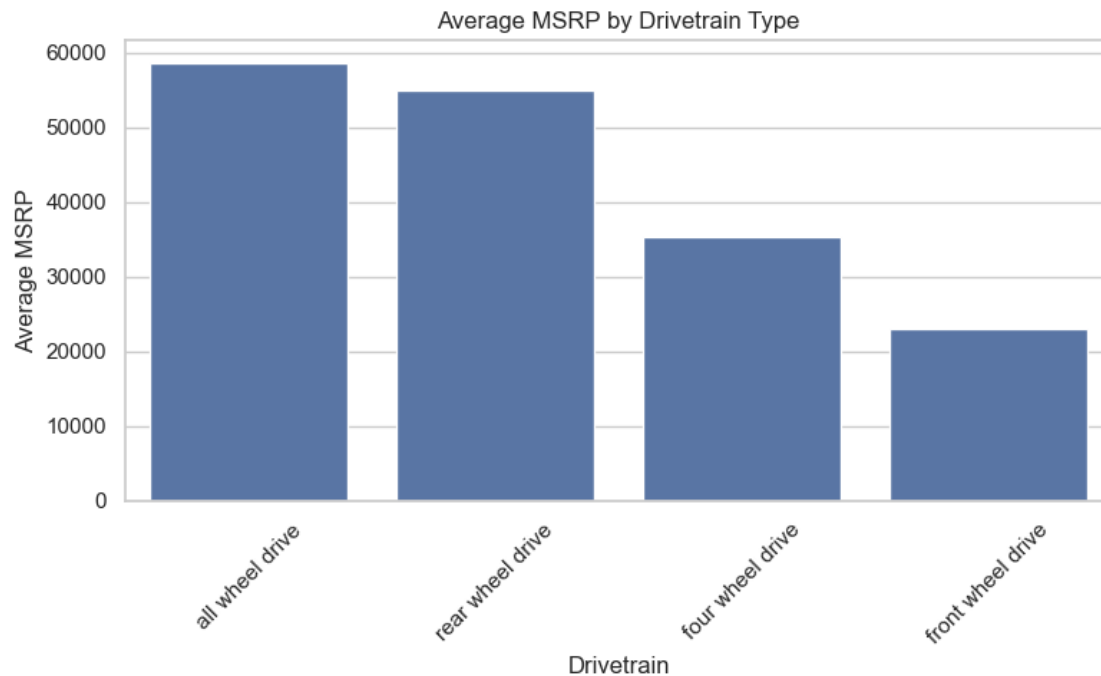
plt.figure(figsize=(10, 6))
sns.boxplot(data=price_subset, x="Driven_Wheels", y="MSRP")
plt.title("Vehicle Price by Drivetrain Type (MSRP <= $100,000)")
plt.xlabel("Drivetrain")
plt.ylabel("MSRP")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



1.1.7 Average MSRP by drivetrain

```
[27]: avg_price = cars.groupby("Driven_Wheels", as_index=False)["MSRP"].mean().
      ↪sort_values("MSRP", ascending=False)

plt.figure(figsize=(8, 5))
sns.barplot(data=avg_price, x="Driven_Wheels", y="MSRP")
plt.title("Average MSRP by Drivetrain Type")
plt.xlabel("Drivetrain")
plt.ylabel("Average MSRP")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



1.1.8 Accidents by season

```
[28]: accidents["Start_Time"] = pd.to_datetime(accidents["Start_Time"],  
      ↪errors="coerce")  
accidents["Month"] = accidents["Start_Time"].dt.month  
  
winter_accidents = accidents[accidents["Month"].isin([12, 1, 2])].copy()  
  
print(winter_accidents.shape)
```

(1997071, 47)

```
[30]: spring_accidents = accidents[accidents["Month"].isin([3, 4, 5])].copy()  
  
print(spring_accidents.shape)
```

(1533186, 47)

```
[31]: summer_accidents = accidents[accidents["Month"].isin([6, 7, 8])].copy()  
  
print(summer_accidents.shape)
```

(1535219, 47)

```
[32]: fall_accidents = accidents[accidents["Month"].isin([9, 10, 11])].copy()
```

```
print(fall_accidents.shape)
```

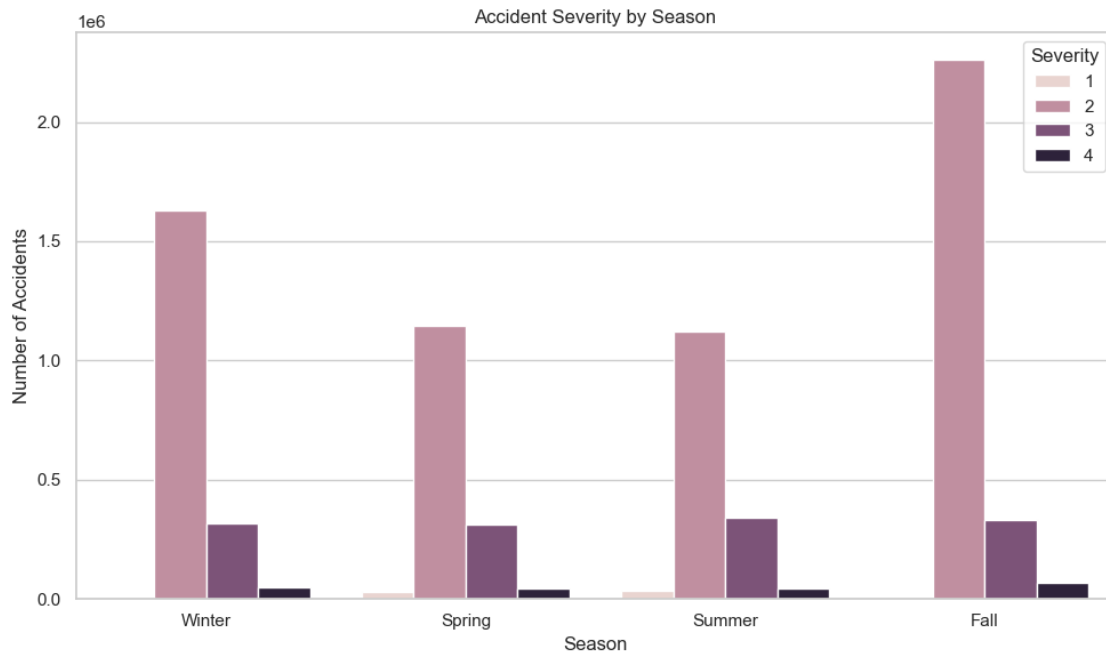
```
(1919752, 47)
```

```
[33]: accidents["Start_Time"] = pd.to_datetime(accidents["Start_Time"],  
      ↪errors="coerce")
```

```
[34]: accidents["Month"] = accidents["Start_Time"].dt.month
```

```
[35]: def get_season(month):  
      if month in [12, 1, 2]:  
          return "Winter"  
      elif month in [3, 4, 5]:  
          return "Spring"  
      elif month in [6, 7, 8]:  
          return "Summer"  
      else:  
          return "Fall"  
  
accidents["Season"] = accidents["Month"].apply(get_season)
```

```
[36]: season_subset = accidents[["Season", "Severity"]].dropna()  
  
plt.figure(figsize=(10, 6))  
sns.countplot(data=season_subset, x="Season", hue="Severity",  
              order=["Winter", "Spring", "Summer", "Fall"])  
plt.title("Accident Severity by Season")  
plt.xlabel("Season")  
plt.ylabel("Number of Accidents")  
plt.tight_layout()  
plt.show()
```



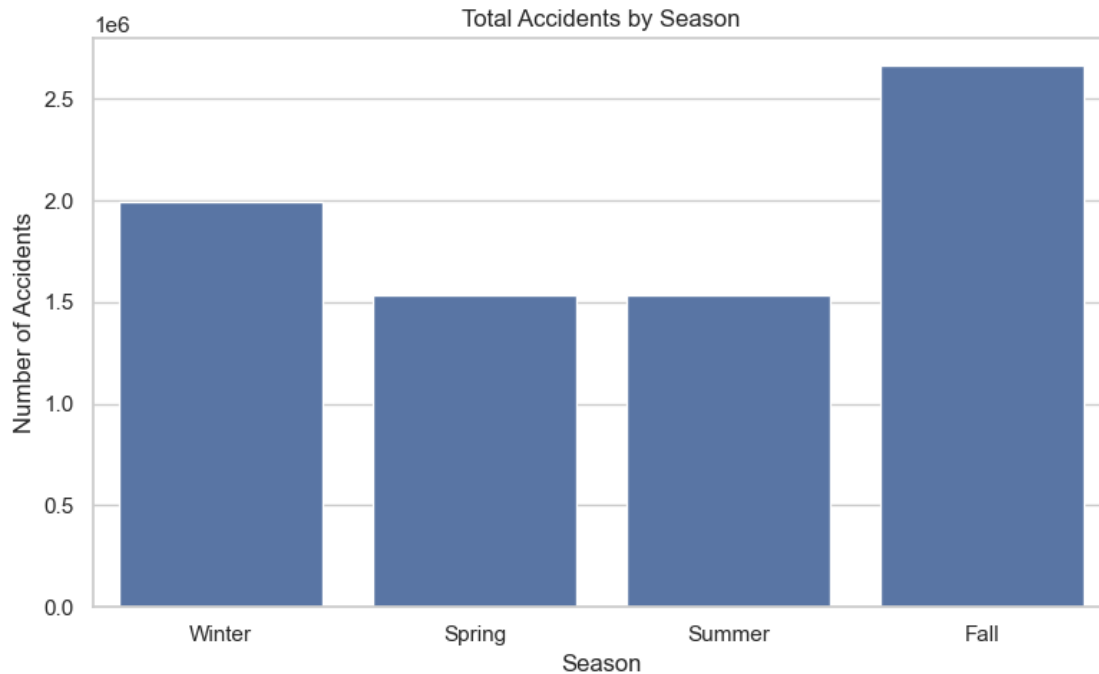
```
[37]: accidents["Season"].value_counts()
```

```
[37]: Season
Fall      2662918
Winter    1997071
Summer    1535219
Spring    1533186
Name: count, dtype: int64
```

```
[38]: accidents.groupby("Season")["Severity"].count()
```

```
[38]: Season
Fall      2662918
Spring    1533186
Summer    1535219
Winter    1997071
Name: Severity, dtype: int64
```

```
[39]: plt.figure(figsize=(8, 5))
sns.countplot(data=accidents, x="Season",
              order=["Winter", "Spring", "Summer", "Fall"])
plt.title("Total Accidents by Season")
plt.xlabel("Season")
plt.ylabel("Number of Accidents")
plt.tight_layout()
plt.show()
```

There are definitely more accidents in the colder months, which isn't really surprising since there is typically less grip due to colder temperatures causing the rubber compounds to harden in tires and not having open pores per say to grip the asphalt/concrete better. Lower temp = less friction. Less friction = less grip. I do think that the pricing of AWD (all wheel drive) is geared towards larger profit margins for manufacturers and is exacerbated by rating systems like the IIHS safety picks. That's also a reason why headlights are so bright now, but that's probably going to be saved for another project :)

[]: